Derivatives: Contracts, Models, and Markets · Technical Note 29

What Hagan's Formula Says: the Level, the Backbone, the Skew, and Where It Fails

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Section 29.9 of Derivatives: Contracts, Models, and Markets sets out SABR, writes Hagan's expansion, states what each parameter controls, and uses the formula to fit a smile; it asserts that β is nearly redundant with ρ and fixes it outside the smile for that reason. It derives none of this, and the derivation usually asked for, the singular-perturbation argument that produces the expansion, is a research paper and teaches a desk nothing it uses. This note derives the results that a desk does use, all of which follow from expanding Hagan's formula itself in log-moneyness. Three lines of algebra give the at-the-money level, the backbone, the skew, and the curvature; the skew turns out to depend on β and ρ only through one combination, which is the redundancy made precise and the reason the calibration recipe fixes β from a time series rather than a cross-section. The curvature, by contrast, separates them. The note closes with the boundary of validity (the expansion is in λk, not k) and with what goes wrong at low strikes, where the formula implies a negative density.

1 THE EXPANSION

Write the smile in **log-moneyness** $k = \log(K/F)$, and abbreviate

$$\sigma_0 = \alpha F^{\beta-1}, \qquad \lambda = \frac{\nu F^{1-\beta}}{\alpha} = \frac{\nu}{\sigma_0} \tag{TN29.1}$$

Both are worth a sentence, because everything below is written in them. σ_0 is the leading at-the-money volatility: it is what Hagan's formula returns at $K = F$ before the $(1 + CT)$ correction. And λ is the **vol-of-vol measured in units of the volatility itself**: dimensionless, and the only way ν ever appears. A desk that reports $\nu = 0.35$ has said nothing until the volatility level is also known; $\lambda = 1.8$ says the volatility of volatility is nearly twice the volatility.

Expanding Hagan's formula (29.22) about $k = 0$ gives

$$\frac{\sigma_B(k)}{\sigma_B(0)} = 1 + r_1 k + r_2 k^2 + O(k^3) \tag{TN29.2}$$

with

$$r_1 = -\frac{1}{2}\Big[(1-\beta) - \rho\lambda\Big] \tag{TN29.3}$$

$$r_2 = \frac{1}{12}\Big[(1-\beta)^2 + (2-3\rho^2)\lambda^2\Big] \tag{TN29.4}$$

These are identities of the formula in the $T \to 0$ limit, not further approximations, and they are the whole content of the note: Sections 2 to 4 are readings of (TN29.3) and (TN29.4).

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The absence of a cross term in (TN29.4) is not an oversight. A term in $\rho\lambda(1-\beta)$ is dimensionally admissible and is exactly what one expects after seeing (TN29.3), where β and ρ are inseparably mixed. It is absent, and its absence is what Section 4 is about.

2 THE LEVEL AND THE BACKBONE

At $k = 0$ the correction factor is all that survives, so

$$\sigma_{\text{ATM}} = \alpha F^{\beta-1}\big(1 + CT\big), \qquad C = \frac{(1-\beta)^2}{24}\frac{\alpha^2}{F^{2-2\beta}} + \frac{\rho\beta\nu\alpha}{4F^{1-\beta}} + \frac{2-3\rho^2}{24}\nu^2 \quad \text{(TN29.5)}$$

which is the expression Section 29.9's calibration recipe inverts to get α from the at-the-money quote.

The backbone is one derivative of this. Ask what happens to at-the-money volatility when the forward moves and the parameters do not. Differentiating $\alpha F^{\beta-1}$,

$$\frac{\partial \sigma_{\text{ATM}}}{\partial F} \approx (\beta - 1)\,\frac{\sigma_{\text{ATM}}}{F} \qquad \text{(TN29.6)}$$

So β does not merely influence the backbone; it is the backbone. At $\beta = 1$ the derivative vanishes: at-the-money volatility does not move when the forward moves, which is what "flat in log F" means and is the lognormal convention. At $\beta = 0$ it is $-\sigma_{\text{ATM}}/F$, exactly the rate at which a constant *normal* volatility must translate into a falling lognormal one, which is flat in F. Everything between interpolates.

Bumping F and refitting nothing measures the derivative directly, and the table sets it against (TN29.6) (the last three rows with $\alpha = 0.20\,F^{1-\beta}$, $\rho = -0.30$, $\nu = 0.35$; $F = 3\%$, $T = 5$ for $\beta = 0$ and $\beta = 1$, $F = 5\%$, $T = 2$ for $\beta = 0.7$):

	Measured	**Equation** (TN29.6)	**Gap**
Section 29.9's fit, $\beta = 0.5$	−2.4783	−2.4995	0.9%
Normal, $\beta = 0$	−7.1277	−7.0166	1.6%
Lognormal, $\beta = 1$	0.0000	0.0000	—
$\beta = 0.7$	−1.2046	−1.2127	0.7%

This is also the measurement a desk actually makes. Section 29.9 says β is chosen "by judgment or by regressing historical at-the-money volatility on the level of the forward." Equation (TN29.6) is that regression written down: in logs it is a straight line of slope $\beta - 1$.

3 THE SKEW, AND WHY β AND ρ CANNOT BE SEPARATED

Equation (TN29.3) says the at-the-money skew has *two additive sources*:

$$\underbrace{-\frac{1}{2}(1-\beta)\,\sigma_0}_{\text{from the CEV exponent}} \quad + \quad \underbrace{\frac{1}{2}\rho\lambda\,\sigma_0}_{\text{from the correlation}} \qquad \text{(TN29.7)}$$

The first is the backbone effect seen across strikes rather than across time: with $\beta < 1$ the proportional volatility $\alpha F^{\beta-1}$ rises as the rate falls, so low strikes carry higher volatility, with no stochastic volatility needed. The second is Section 21.4's leverage mechanism: with $\rho < 0$ volatility rises as the rate falls, which again makes

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low strikes expensive. On an equity surface both terms are negative and reinforce; on a rate surface where ρ is sometimes positive they can oppose.

The point follows. β and ρ appear in (TN29.3) only through the single number

$$\Lambda = (1 - \beta) - \rho\lambda \tag{TN29.8}$$

A smile measures Λ. It does not measure β and ρ separately, because to first order in k nothing in the data distinguishes a pair from any other pair with the same Λ. *This is Section 29.9's "nearly redundant with ρ," stated exactly, and it is a fact about the model rather than about the optimizer.*

The qualification is in the word *nearly*. The redundancy is exact only at first order, and the second order does separate them—so the question is how much of the smile lives there. Taking Section 29.9's fitted smile ($F = 4\%$, $\beta = 0.5$, $\alpha = 0.0387$, $\rho = -0.30$, $\nu = 0.35$, $\lambda = 1.81$, $\Lambda = 1.0426$), holding λ and Λ fixed, re-solving for ρ at each β, and re-solving α so that the at-the-money volatility stays at 20.00% (so $\nu = \lambda\sigma_0$ moves slightly, from 0.342 to 0.371):

β	α	ρ	**Implied volatility (%) at strike**						**Worst**
			2.5%	3.0%	4.0%	4.5%	5.0%	5.5%	**gap**
0.50	0.0387	−0.300	26.25	23.60	20.00	18.92	18.26	17.91	—
0.00	0.0076	−0.024	27.03	23.92	20.00	18.94	18.31	17.98	0.78
0.25	0.0171	−0.162	26.65	23.76	20.00	18.93	18.30	17.98	0.40
0.75	0.0886	−0.438	25.86	23.43	20.00	18.90	18.18	17.78	0.39
1.00	0.2053	−0.576	25.50	23.29	20.00	18.86	18.05	17.54	0.75

Across β from 0 to 1 the whole smile moves by less than one volatility point, which on the wings of a swaption grid is inside the bid–ask. The same sweep *without* compensating (holding ρ at -0.30, with λ held and α still re-solved from the at-the-money quote) costs 2.66 points at $\beta = 0$ and 2.31 at $\beta = 1$. So compensating through Λ removes about two-thirds of the discrepancy, and what remains is second-order and sits in the low-strike wing.

Three consequences follow, and they are the practical content of the note.

Fitting β and ρ together on one smile is fitting a ridge. The objective function is nearly flat along the curve Λ = constant, so the optimizer's answer is set by its starting point and by rounding. Refit tomorrow and it lands elsewhere on the ridge, with the fit quality unchanged and both parameters moved. This is the phantom profit and loss of Section 21.10, generated by a model that has not changed its mind about anything the market can see.

The escape is to measure β where it is not degenerate. Equation (TN29.6) identifies β from the *time series* of at-the-money volatility against the forward, which is a different observation from the cross-section that only sees Λ. This is why Section 29.9's recipe fixes β first and from history, and why doing so is not an admission of defeat.

And the residual is where the choice of β is paid for. Since the wings are what separate the pairs, a book concentrated in the wings (which is where structured rate products sit) is the one book for which β genuinely matters, and the one for which the sub-point gaps in the table above are not small enough to ignore.

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4 THE CURVATURE

Equation (TN29.4) says the curvature is

$$r_2 = \frac{1}{12}\left[(1-\beta)^2 + (2-3\rho^2)\lambda^2\right]$$

and three things follow.

ν produces the curvature, and λ^2 is how. With β and ρ held, the stochastic-volatility part of the curvature is quadratic in the vol-of-vol, so doubling ν quadruples that part and, at the fitted parameters, nearly quadruples the smile's convexity. This is Section 29.9's statement that ν produces the curvature, made quantitative.

β and ρ are *not* redundant here. There is no $\rho\lambda(1-\beta)$ term: the two enter the curvature through $(1-\beta)^2$ and $\rho^2\lambda^2$ separately, and both contributions are even. So the second order is exactly where the pair becomes identified, which is the analytical statement of what Section 3's table showed numerically, and it explains why the residual there was concentrated in the wings rather than spread across the smile.

The coefficient $2-3\rho^2$ changes sign. At $|\rho| > \sqrt{2/3} \approx 0.8165$ the stochastic-volatility contribution to curvature is *negative*, and with $\beta = 1$, where the first term vanishes, the smile is concave rather than convex. A strongly correlated SABR does not smile; it bends the other way. This is the same phenomenon Table 21.2 records for Heston, whose curvature at $\rho = -0.9$ is -0.58 against $+0.36$ in the base case, and the agreement is not a coincidence: both models are producing skew by correlating volatility with price, and past a point the tilt overwhelms the convexity.

5 WHERE THE EXPANSION STOPS WORKING

The expansion is in λk, not in k. This is visible in Hagan's own formula, whose intermediate variable is $z = -\lambda k$ to leading order; k is never small on its own terms. The quadratic (TN29.2) is therefore good while $|\lambda k|$ is well below 1, and not otherwise, and the difference is stark (the steep smile has $F = 3\%$, $T = 1$, $\alpha = 0.01$, $\beta = 0.5$, $\rho = -0.5$, $\nu = 0.6$):

	$k = \log(K/F)$				
	−0.20	−0.10	0	0.10	0.20
Section 29.9's fit ($\lambda = 1.81$): Hagan	22.40%	21.12%	20.00%	19.07%	18.38%
quadratic (TN29.2)	22.47%	21.13%	20.00%	19.06%	18.31%
A steep short-dated smile ($\lambda = 10.4$): Hagan	9.96%	7.86%	5.87%	5.24%	6.20%
quadratic (TN29.2)	11.86%	8.20%	5.87%	4.86%	5.17%

At $\lambda = 1.8$ the quadratic tracks the formula to within seven basis points of volatility across a ±20% strike range. At $\lambda = 10$ it is nearly two volatility points out at the same moneyness, because $\lambda k = 2$ there. A low-volatility, high-vol-of-vol environment (which is where rate markets spent the 2010s) is exactly where the readings above should be used qualitatively and not as formulas.

Low strikes, and the negative density. Section 29.9 records that Hagan's expansion can imply a negative density at low strikes. The mechanism is visible in the formula.

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As $K \to 0$ with $\beta < 1$, the factor $(FK)^{-(1-\beta)/2}$ diverges while $z \to 0$ and $z/x(z) \to 1$; $D(K)$ grows like $\log^4(F/K)$, and the correction factor is dominated by the first term of C, which grows like $(FK)^{-(1-\beta)}$, so

$$\sigma_B(K) \;\sim\; \frac{80\,\alpha^3 T}{(1-\beta)^2} \cdot \frac{(FK)^{-3(1-\beta)/2}}{\log^4(F/K)} \tag{TN29.9}$$

which is unbounded. An unbounded implied volatility drives the Black price of a low-strike call to the forward, and a call price that stops rising as the strike falls has a negative second derivative, and by Breeden–Litzenberger that second derivative is the density. Nothing in the true SABR process does this; the process has an absorbing boundary at zero that the asymptotic formula never sees.

Where it bites, for Section 29.9's fitted smile and variations on it (the β rows rescale α to hold σ_0):

	Density first negative below	**As a fraction of F**
Section 29.9's fit ($\nu = 0.35$, $\beta = 0.5$, $\rho = -0.30$)	0.128%	0.03
$\nu = 0.60$	0.982%	0.25
$\nu = 0.20$	none found	—
$\beta = 0$	0.538%	0.13
$\beta = 1$	none found	—
$\rho = -0.70$	0.114%	0.03

Two readings follow, and the second is the one that matters. At the fitted parameters the failure is at a strike of about thirteen basis points—far outside anything quoted, and safely ignorable. Raise the vol-of-vol to $\nu = 0.60$ and it moves to a quarter of the forward, which against a 4% rate is a strike of 1%: a strike that trades. *So whether the arbitrage is academic or real is a property of the calibration, not of the model*, and a desk that has never checked is a desk that does not know which case it is in. At $\beta = 1$ it does not arise at all, which is one of the few genuine arguments for the lognormal choice.

The repairs are named in Section 29.9: shifting F and K away from zero, or replacing the expansion with a numerical solution of the model's own forward equation—Hagan's later "arbitrage-free SABR." Both are in use at dealers, and (TN29.9) is why one of them is needed.

6 WHERE THIS CAME FROM

Hagan, Kumar, Lesniewski, and Woodward's "Managing Smile Risk" (2002) is the source of the expansion, and its appendix contains the singular-perturbation argument this note deliberately does not reproduce. The reading of the expansion given here—the level, the backbone, the skew decomposition and the Λ degeneracy—is closest in spirit to Rebonato, McKay and White's *The SABR/LIBOR Market Model* (2009), which sets out why calibration in the presence of a flat direction is a practical problem rather than a numerical one. West (2005) documents how the fitted parameters actually move day to day when the ridge is not respected. Hagan, Kumar, Lesniewski, and Woodward returned to the arbitrage of Section 5 in "Arbitrage-Free SABR" (2014), which replaces

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the expansion with a one-dimensional partial differential equation; Antonov, Konikov, and Spector (2013) give exact results for the zero-correlation case, which is the other way of bounding what the expansion is doing wrong. Andersen and Piterbarg's second volume is the reference for shifted and free-boundary variants.

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